Smoke Rescue

Two robot dogs search a burning house, share one map, and walk back out.

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When a house is on fire, the first crew inside can see about an arm's length ahead. Every minute spent searching a room twice is a minute a trapped person does not have. We asked one question: can two small robots go in first, split the rooms, find the people, and come back out, with nobody steering them?

Smoke Rescue is our answer, built and tested in simulation. Two quadruped robots enter a burning house through the front door with no floor plan. Each robot maps the house with its own LiDAR and depth camera and confirms people with an OpenCV 5 person detector. It shares its map with its partner over a lossy radio. OpenCV 5 ORB place recognition lines up the two maps when both robots see the same place. On 39 real floor plans the brain had never seen, two robots found 99 % of people and one robot found 92 % ($p = 0.002$). Neither set-up reported a single false person.

What this report covers

§	Section	Read it to learn
1	Problem	Why a burning house is hard for a robot team
2	Users	Who this is for, and what they get from it
3	Architecture	How the brain, the radio and the agent fit together (diagram)
4	OpenCV 5 implementation	Person detection, thermal, ORB map linking and the 3D map, with measurements
5	AWS deployment	What runs on AWS and how a live mission starts
6	Evaluation	The 39-house benchmark, the three filmed missions and the agent evidence
7	Limitations	What does not work well yet
8	Responsible use	The safeguards, and what we do not claim

Short on time? Read section 4 and the first table in section 6. Build steps, pinned versions and deploy commands are in the GitHub README: https://github.com/TashonBraganca/smoke-rescue. The live judge website is https://main.dyhvyjxo1bz4g.amplifyapp.com.

1. Problem

Smoke takes away sight. Heat takes away time. Two searchers who cannot share what they saw will search the same rooms twice. A robot team for this job has to answer four questions with its own sensors:

- Where am I, and where is my partner?
- Which rooms have we already searched?
- Is that shape a person?
- Can we both get back out?

We kept the limits that a real team would face. No robot gets a floor plan. The only shared reference is the launch pad at the front door. The radio drops messages, has a short range and carries at most 8 KB per message. Each robot sees only through its own RGB-D camera, thermal camera and planar LiDAR.

2. Users

We built Smoke Rescue for two groups of people.

- **Fire incident commanders.** Before a crew goes in, the commander sees one map with the searched rooms, the people found and the fire. The crew can go straight to a confirmed person instead of searching room by room.
- **Robotics teams.** Engineers can test a multi-robot search policy on hundreds of real floor plans, scored against hidden ground truth, before they put it on hardware.

The judge website shows both views. Three houses play a recorded mission with each robot's camera. Two houses start a live mission on AWS that anyone can watch as it runs.

3. Architecture

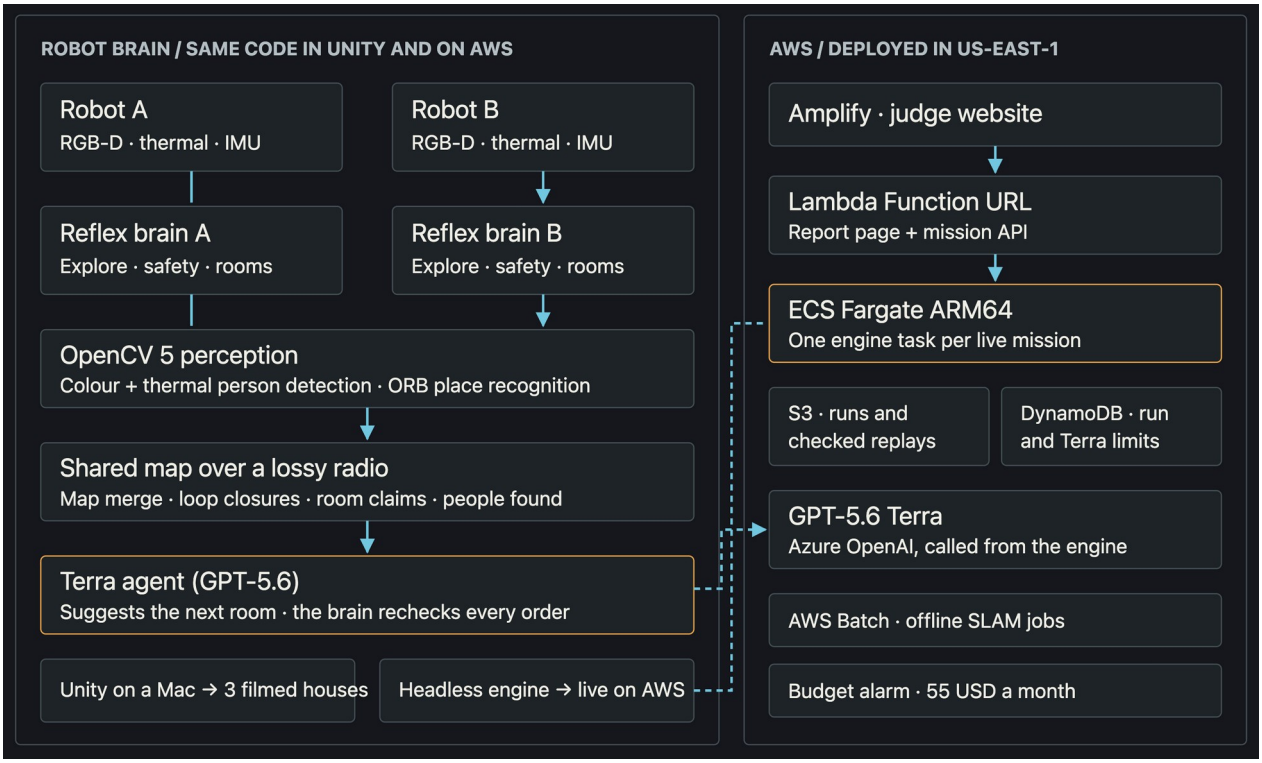


Figure 1. The robot brain (left) and the AWS deployment (right). The same brain code runs in Unity for the films and on AWS for live missions.

Each robot runs the same brain, written in C#. It runs at 60 Hz and keeps the robot safe with no language model in the loop. The brain runs in two places. In Unity HDRP on a Mac it drives the three filmed houses, with rendered cameras. In a headless simulator on AWS it drives the live missions, without rendered images.

The brain runs a mission in five steps:

1. **Enter.** The robot walks straight in along its launch line to 1.5 m past the door before it picks any other goal.
2. **Search.** LiDAR and depth build an occupancy grid. A room ledger ranks the next goal. Door claims stop both robots from taking the same doorway.

3. **Confirm people.** An OpenCV 5 detection on the floor within 4.5 m confirms a person. A farther detection or a warm shape becomes a candidate that the robot walks up to and checks.
4. **Keep clear of fire.** Hot air marks a room. Measured fire gets a 1.2 m margin.
5. **Walk out.** At the search deadline, or when no unentered room remains, both robots return to the door. They use a planned route, or their own recorded trail if no route exists.

Shared context. Map layers, room records, door claims, people found and ORB keyframes travel over the lossy radio. A robot converts partner data through the measured link between the two maps, never through ground truth.

Agent. Terra, a GPT-5.6 agent, reads a robot's report a few times per mission and suggests the next room to search. It sees only that robot's pose, the offered frontiers, a map summary and the radio inbox. In Unity agent runs it also calls the OpenCV tools and must cite an evidence id for each risky order. Python guardrails check each order, then the brain checks it again and sends any refusal back to the model. The three films run the brain alone, so every decision in them belongs to the brain.

4. OpenCV 5 implementation

Only an OpenCV 5 detection confirms a person. A warm shape on the thermal camera is a reason to go and look. It is never a confirmation on its own.

Component	OpenCV 5 call	What it decides	Measured
Person detector	cv2.dnn with YOLO11m (ONNX). Range comes from the robot's own depth image.	Is a person there, and where	3,378 Unity frames at confidence 0.5: recall 0.51 per visible person view (86 of 170), precision 0.99 (1 false box), median range error 0.30 m. 1,266 ms per frame on an M5 CPU.
Thermal	Thresholds, contours, morphology	Fire (200 °C and above) and body heat (28 to 40 °C)	Sets the fire margins and the person candidates
Place recognition	ORB (200 features with depth), BFMatcher Hamming kNN, ratio test 0.85, RANSAC rigid fit on 3D points	Which partner view shows a place I mapped, and how our two maps line up	0 false accepts in about 30,000 negative pairs. Cross-robot recall 11.7 % per keyframe pair. 7,605 bytes per keyframe.
3D map	cv2 HashTSDF volume per robot	Obstacle tops for the planner	About 1.04 million surface points per robot per mission
Agent tools	Laplacian variance, dark channel, contours	assess_visibility, inspect_candidate, scan_thermal, track_fire_spread, measure_opening	assess_visibility takes 8.6 ms on Fargate ARM64

Two robots, one map

The launch pads give both robots a shared frame. A pad set slightly wrong still turns a robot's whole map. In the one-door house, robot B started 2.39 degrees off its pad, which puts its far walls about 0.6 m out over 15 m. That is enough to draw a door inside a wall.

Each robot stores an ORB keyframe with depth every 2 s and sends it over the radio (7.6 KB). A robot compares a partner keyframe only with its own views near the place where the partner should be, within 3 m and 60 degrees. A match must pass RANSAC with at least 10 inliers. The robot then checks it a second way: the partner's LiDAR sweep from that moment must fit the robot's own LiDAR map. Two matches that agree, taken at least 1 m apart, set the link.

- The camera link measured robot B's start tilt as 2.42 degrees against a true 2.39 degrees, with no duplicated people.
- Wall agreement between the two maps rose from 72 % with the pads alone to 97 % with the camera link.
- The best run verified 138 loop closures. Robot A's median drift was 3.1 cm and 0.25 degrees.

We tried cv2.Odometry as the pose source and dropped it. It drifted 17 degrees in 4 s on our frames. The brain uses gyro heading and corrects it with LiDAR scan matching.

5. AWS deployment

Everything a judge touches runs on AWS in us-east-1 (right side of Figure 1).

- **Amplify** hosts the judge website.
- **A Lambda Function URL** serves the mission API and the report page.
- **ECS Fargate ARM64** starts one engine task per live mission. The task runs the same C# brain in the headless simulator and writes the mission to S3 as it runs.
- **GPT-5.6 Terra on Azure OpenAI** answers the agent calls from the engine task. DynamoDB counts them: at most 20 per mission and 100 per day.
- **A Fargate ARM64 service** (1 vCPU, 2 GB) runs the agent and OpenCV 5 API.
- **AWS Batch** runs offline SLAM jobs. A budget alarm caps spend at 55 USD a month.

How a live mission runs:

1. The judge picks Bungalow or Corner house and presses "Send them in".
2. A five-second setup screen shows the house while the API starts a Fargate task. The first frames arrive in about 30 s.
3. The page polls the mission API every 1.5 s and draws the 3D house, the shared map and the team radio from the streamed frames. Terra's decisions appear in the radio as they happen.
4. At most two live missions run at once. If AWS cannot start one, the page plays a checked recording of the same setup and labels it. A recording counts as checked only if the robots entered every room, both got out, and nobody reported a false person.

We checked the whole path on 27 October. A Corner house mission started, streamed live, and showed Terra's room choices in the team radio.

6. Evaluation

Every score comes from ground truth that no robot-side code can read.

6.1 Benchmark: 39 floor plans the brain never saw

HouseExpo floor plans, headless simulator, one mission per house per set-up, 25 October. Brackets are 95 % bootstrap intervals. P values come from Wilcoxon tests on per-house pairs.

Measure	Two robots, shared map	One robot	Difference
People found	99.0 % [97.4, 100]	92.3 % [86.7, 96.4]	+6.7 points, p = 0.002
Rooms entered	99.2 %	93.9 %	+5.3 points, p = 0.002
Mean time to confirm a person	67.2 s	94.5 s	27.3 s sooner
False confirmations	0	0	none
Every robot back outside	60 % of missions	90 % of missions	30 points worse, p =

			0.003

Two robots find more people, sooner. They also get every body back outside less often, and we report that plainly. We also ran 120 missions on 30 generated houses. There, the radio cut the time to confirm a person from 134.0 s to 96.6 s. It also cut repeated floor from 77 % to 63 %.

6.2 The three filmed missions

Unity HDRP, brain only, scored against hidden ground truth.

House	People found	False	Both robots out	Mission time
Three-bed ranch	5 of 5	0	Yes, at 365 s and 375 s	6:16
L-shaped ranch	4 of 5	0	Yes, at 369 s and 413 s	6:53
One-door house	4 of 5	0	Yes, along their own trail, at 454 s and 463 s	7:44

6.3 Agent evidence

34 recorded agent runs in Unity.

Measure	Value
Agent steps / tool calls	1,628 / 6,403
OpenCV 5 perception calls	2,727
Robot orders that cite an evidence id	990 of 1,571 (63 %). The rest need no evidence, for example a turn or a team message.
Cited ids that match a recorded OpenCV call	824 of 863
Guardrail refusals	169, most often a confirmation from beyond 4 m (29)

6.4 How we test a change

We trace every failed take to a cause in its own logs before we change code. Each fix ships with a test that fails without it, and the brain's logic suite passes 305 of 305. Two of our worst setbacks came from our own changes: an early exit rule and a narrower confirm range. Both showed up in the next recording. Since then, no change ships without a recording audit.

7. Limitations

- **One look is not enough.** A single camera view finds a visible person 51 % of the time, so the robot has to come close and look again. One view misses people who are far away, behind furniture or in thick smoke.
- **Getting out is the weakest part.** Every robot came back outside in 60 % of two-robot benchmark missions. Our 26 October fixes target this. All three films end with both robots outside, but three films are not a benchmark.
- **Map linking is sparse.** Only 11.7 % of true same-place pairs match across robots. Robots that never share a view do not link their maps.
- **Live missions have no camera picture.** AWS runs the brain without rendering, so the live view draws positions, maps and finds, not video.

8. Responsible use

A system that tells a crew "nobody is in that room" can cost a life if it is wrong. We designed around that one failure.

- **Heat alone never counts.** A robot reports a person only after an OpenCV detection that it checks up close. False confirmations stayed at 0 in the benchmark and in all three films.
- **The language model advises. It does not decide.** Terra cannot override the safety code. Every risky order needs evidence that it can cite, and the brain rejects anything else and tells the model why.
- **We score honestly.** Unity assembly definitions and a pytest check keep ground truth away from robot code. No brain setting names a house. We wrote down every change to the film houses, such as a slower fire and two beds turned against a wall, with its reason.
- **A real fire needs more than this.** Before this goes near one, it needs hardware trials, a validated fire model and trained responders in the loop.
- **We cap cloud spend.** A 55 USD budget alarm, request rate limits and model routes closed to the public keep a busy day from turning into a large bill.

Reproduce it: build steps, pinned versions and deploy commands are in the README at https://github.com/TashonBraganca/smoke-rescue.